POLICY BRIEF

A JOB? IN THIS ECONOMY? - Understanding State-Level Variation in Female Labour Force Participation in India

An exploratory state-level analysis of economic, educational, structural, social and demographic correlates

Project feature	Description
Research question	Which state-level factors are most strongly associated with female labour force participation?
Primary outcome	Female LFPR, PLFS 2021–22
Analytical sample	32 complete State/UT observations in the combined model
Approach	Correlation and multiple linear regression; exploratory, non-causal

Prepared by Sivachanthini Ravikumar | September 2026

This brief translates an independent R-based state-level analysis into a policy-relevant research product. It does not establish causal effects.

Executive Summary

The policy problem. Female labour force participation (LFPR) is a central indicator of women's economic inclusion, yet national averages conceal substantial differences across Indian states and Union Territories. The policy challenge is therefore not simply to raise participation, but to understand the economic, structural and social conditions associated with where participation is higher or lower.

What this study examines. The analysis examines six state-level variables: secondary education and above, NSDP, urbanisation, female non-agricultural employment share, women married before age 18, and total fertility rate (TFR). Three regression specifications separate economic/structural factors, social/demographic factors, and a combined model.

What the analysis finds. The combined model explains 40.5% of the observed cross-state variation in female LFPR (adjusted $R^2 = 0.262$; overall $p = 0.030$). However, none of the six individual predictors is statistically significant at the 5% level in the combined model. Urbanisation has the largest absolute standardized coefficient ($\beta = -0.457$; $p = 0.145$), but the estimate remains statistically uncertain.

Policy implication. The results do not support a single-factor explanation of women's labour-force participation. They instead support a multidimensional policy lens combining employment opportunities, education-to-work transitions, safe mobility, care infrastructure and attention to social and demographic constraints.

At a Glance

Indicator	Figure	Interpretation
India female LFPR, PLFS 2021–22	32.8%	Women aged 15+; usual status (PS+SS), rural + urban
Rural female LFPR	36.6%	Higher than the urban rate in the same PLFS measure
Urban female LFPR	23.8%	Below the rural rate in 2021–22
India female LFPR, PLFS 2023–24	41.7%	Later national benchmark; not the outcome period used in the state analysis
Combined model R^2	0.405	40.5% of observed state-level outcome variation explained
Combined model adjusted R^2	0.262	Fit after accounting for six predictors

Source / note: PLFS figures: MoSPI/PIB, Annual PLFS 2021–22 and 2023–24. Regression figures: author's analysis.

1. The Policy Problem

Female LFPR is not the same as employment. LFPR captures the share of the relevant population that is in the labour force—working or seeking/available for work—and therefore differs from the worker population ratio or employment rate. A rise in LFPR is important, but it does not by itself reveal whether women have access to secure, adequately paid, productive or preferred work.

In PLFS 2021–22, the all-India female LFPR for persons aged 15 years and above under usual status (principal plus subsidiary status) was 32.8%. The rural rate was 36.6%, compared with 23.8% in urban areas. By PLFS 2023–24, the national female LFPR had reached 41.7% under the same broad measure. These figures demonstrate both the scale of change over time and the importance of distinguishing national trends from state-level variation.

Source / note: PLFS 2021–22 and 2023–24, Government of India.

The policy question is therefore not simply whether female participation is increasing nationally, but what conditions are associated with differences between states. Economic structure, education, urbanisation, the availability of non-agricultural work, marriage timing and fertility represent distinct dimensions of that policy environment.

2. Research Question and Objectives

Primary research question. Which economic, educational, structural, social and demographic factors are most strongly associated with female labour force participation across Indian States and Union Territories?

- Examine associations between female LFPR and educational attainment, economic conditions, urbanisation and employment structure.
- Examine associations between female LFPR and early marriage and total fertility rate.
- Compare the explanatory performance of economic/structural, social/demographic and combined specifications.
- Translate observed patterns into policy considerations while distinguishing association from causation.

3. Analytical Framework

The study treats female LFPR as the outcome and groups the explanatory variables into two dimensions.

Dimension	Variables	Policy logic
Economic / structural	Secondary education and above; NSDP; urbanisation; female non-agricultural employment share	Captures human capital, economic conditions, spatial structure and employment composition.
Social / demographic	Women married before age 18; total fertility rate	Captures life-course and demographic context associated with women's labour-market participation.

These variables are not assumed to represent complete causal mechanisms. For example, urbanisation may simultaneously reflect job access, commuting patterns, living costs, care infrastructure and sectoral composition. The dataset does not allow these mechanisms to be separated.

4. Data and Methodology

4.1 Data sources

Data component	Source / period	Role
Female LFPR	PLFS 2021–22	Dependent variable
Education, NSDP, urbanisation, female non-agricultural employment share	State-level source tables used in the project dataset	Economic / structural predictors
Early marriage	NFHS-5, 2019–21	Social-demographic predictor
Total fertility rate	NFHS-5, 2019–21	Social-demographic predictor

Source / note: NFHS-5 defines the early-marriage indicator as women aged 20–24 married before age 18 and reports TFR in children per woman. Exact source-table metadata are documented in the accompanying variable dictionary.

4.2 Study design

The study uses a state/Union Territory-level cross-sectional design. The unit of analysis is the State/UT rather than the individual woman. Andaman & Nicobar Islands was excluded from the project dataset, and the combined model uses 32 complete observations.

4.3 Statistical approach

Method	Purpose
Bivariate correlations	Assess pairwise linear association between female LFPR and individual predictors.
Multiple linear regression	Estimate associations while considering multiple predictors simultaneously.
Standardized coefficients	Place predictors on a common scale to compare coefficient magnitudes.
Variance inflation factor (VIF)	Assess overlap among predictors that may make individual

	estimates less stable.
Diagnostic analysis	Inspect residual behaviour and potentially influential observations.

4.4 Model specifications

Model	Specification	Purpose
Model 1	Education + log(NSDP) + urbanisation + female non-agricultural share	Economic and structural factors
Model 2	Early marriage + TFR	Social and demographic factors
Model 3	All six predictors	Combined specification

5. Findings

5.1 Model comparison

Model	R^2	Adjusted R^2	Overall p	Interpretation
Model 1	0.297	0.193	0.043	Economic/structural specification has significant overall fit; individual coefficients are uncertain at 5%.
Model 2	0.063	0.002	0.367	Social/demographic specification explains relatively little variation in this sample.
Model 3	0.405	0.262	0.030	Combined specification explains more observed variation and is significant overall.

Model 3 has the highest explanatory power of the three specifications. This should not be interpreted as proof that the added social-demographic variables cause higher or lower LFPR; the models are exploratory and use a small cross-sectional sample.

5.2 Combined model: individual predictors

Predictor	Std. β	p-value	Interpretation
Secondary education and above	+0.021	0.944	Very small estimated association in the combined specification.
Log NSDP	−0.165	0.402	Statistically uncertain.
Urbanisation	−0.457	0.145	Largest absolute standardized coefficient; statistically uncertain.
Female non-agricultural share	−0.245	0.434	Statistically uncertain.
Women married before age 18	−0.282	0.272	Negative estimate; statistically uncertain.
Total fertility rate	−0.288	0.121	Negative estimate; statistically uncertain.

Source / note: Author's regression analysis. Standardized coefficients are from Model 3 (n = 32). None of the six individual coefficients is statistically significant at the 5% level.

5.3 Interpretation

Urbanisation. Urbanisation has the largest absolute standardized coefficient in Model 3 (β = −0.457), but its p-value of 0.145 means that the estimate is not statistically distinguishable from zero at the conventional 5% threshold. It should therefore be described as the largest estimated coefficient in this specification—not as a statistically established determinant.

Social-demographic variables. Early marriage and TFR both have negative estimated coefficients in the combined model, but neither reaches conventional statistical significance. The analysis therefore does not establish an independent negative effect of either variable on female LFPR.

Predictor overlap. VIF values range from 1.35 to 3.98. No predictor crosses a commonly used severe-multicollinearity threshold of 5, but urbanisation and female non-agricultural employment share have VIFs close to 4, consistent with meaningful overlap between these structural measures.

6. Policy Implications

Move beyond single-factor explanations. The results do not support a policy strategy centred on one variable alone. Women's participation should be approached as an outcome of interacting economic, labour-market, educational and social conditions.

Connect education to employment. Education policy should be paired with credible transitions into paid work through skills, apprenticeships, career guidance, digital access and employer linkages.

Treat employment access as a local economic question. State-specific strategies should examine sectoral job availability, commuting constraints, workplace safety, wages and the accessibility of non-agricultural employment rather than assuming that aggregate economic development automatically creates suitable opportunities for women.

Invest in enabling infrastructure. Safe and affordable mobility, childcare and workplace infrastructure can reduce practical barriers to sustained labour-market participation. These mechanisms were not directly measured in the model and therefore warrant further study.

Integrate life-course policy. Policies that support girls' continued education, delay early marriage and strengthen reproductive autonomy should be considered alongside economic inclusion. The objective should be women's agency and opportunity, not fertility reduction as an end in itself.

Build stronger subnational evidence. Future policy analysis should combine labour-market, household, care, mobility and social indicators at district level and, where possible, across multiple years.

7. Recommended Policy Priorities

Priority	Policy direction	Action focus
1	State-specific female employment diagnostics	Establish regular state and district labour-market diagnostics identifying sectors, locations and groups with low participation and limited access to suitable work.
2	Education-to-work pathways for women	Expand industry-linked training, apprenticeships, career counselling and transition support for young women leaving education.
3	Enabling infrastructure for women's work	Strengthen safe transport, childcare, workplace safety and flexible-work arrangements where feasible, particularly around employment clusters.
4	Integrated social and economic policy	Link interventions on adolescent education, early marriage, reproductive autonomy and care responsibilities with women's economic opportunities.
5	Better data for policy targeting	Improve interoperability of state- and district-level datasets covering participation, employment quality, care, mobility and social conditions.

8. Limitations and Responsible Interpretation

- Cross-sectional design: the analysis identifies statistical associations, not causal effects.
- Small sample: the combined model uses 32 complete State/UT observations and six predictors, limiting statistical power and precision.
- Ecological inference: state-level relationships cannot be directly interpreted as relationships between individual women's characteristics and their labour-market behaviour.
- Different reference periods: the primary labour outcome is from PLFS 2021–22, while early-marriage and fertility indicators come from NFHS-5 (2019–21).
- Omitted variables: the model does not directly measure childcare, safety, mobility, wages, household income, social norms, job quality or local labour demand.
- Proxy measures: NSDP, urbanisation and non-agricultural employment share capture broad conditions rather than single mechanisms.
- Model uncertainty: the overall combined model is statistically significant, but no individual predictor is statistically significant at the 5% level.

Appropriate use. The study is best understood as an exploratory evidence map. It identifies patterns that merit policy attention and further investigation rather than providing a ranking of interventions or a causal estimate of policy effects.

9. Conclusion

Female labour force participation varies across Indian states within a policy environment shaped by economic structure, education, urbanisation and social-demographic conditions. In this analysis, the combined specification accounts for 40.5% of observed state-level variation in female LFPR, yet the individual predictors remain statistically uncertain.

The central policy implication is not that one factor should be treated as the decisive lever. Rather, the findings support a multidimensional approach that connects women's human capital to actual employment opportunities while addressing practical and social constraints. Stronger conclusions will require larger samples, more consistent time periods, district-level analysis and richer measures of employment quality and household constraints.

10. Sources

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3. Ministry of Health and Family Welfare (MoHFW). National Family Health Survey (NFHS-5), 2019–21: India Key Indicators and state fact sheets. Government of India.
4. Author's analysis. State-level dataset and R analysis, September 2026; accompanying R script and policy variable dictionary document data preparation and model specifications.

Official source portals: MoSPI PLFS: https://www.mospi.gov.in/PLFS-July-2021-June-2022 | MoHFW reports: https://www.mohfw.gov.in/documents/reports